Research Statement — Amir Hossein Osooli

PhD position: Contextual Integrity for Autonomous AI Agents, ELLIS Institute Tübingen / MPI-IS

The problem I want to work on:

Most current work on agent privacy evaluates single decisions: given this context, should this flow be permitted? I am more interested in the regime where the context itself moves. An agent that behaved correctly in week one holds memories written under permissions that have since changed, relationships that have since ended, and inferences that were never explicitly authorized. The flow it permits in week six may be indefensible even though no single step was wrong.

This connects directly to the argument in AI Agents May Always Fall for Prompt Injections: if an adversary can always construct a context under which a blocked flow appears legitimate, and a defender who tightens norms blocks legitimate flows, then the interesting quantity is not whether a defense holds but where its operating point sits on the curve between over-refusal and leakage. That is a measurement problem, and measurement is what I have been trained to do.

Two projects I would want to start with:

Authorization drift over long horizons. Build a simulated multi-week environment where agent memory is not a flat text store but a set of entries carrying provenance, who disclosed this, to whom, for what purpose, under whose authority. Roles, team membership, and permissions then change during the simulation. Because the environment is generated, the correct norm is known at every step, so violation rate and task utility can be traced as functions of the rate of contextual change rather than reported as single aggregate scores. The question I would want answered: does violation rate rise smoothly with contextual churn, or is there a threshold past which an agent's accumulated memory becomes systematically unsafe? Existing CI benchmarks such as ConfAIde establish that models struggle with tiered contextual reasoning in a static setting; this asks what happens when the tiers themselves are unstable.

Composition and aggregation. Individually harmless disclosures that become sensitive only when combined across turns, across channels, or across agents holding different pieces. In a network where agents represent different users and hold graded rather than uniform trust in one another, I would want to measure whether coordination systematically erodes boundaries, and whether the erosion is driven by adversarial agents or emerges from ordinary cooperative behavior. ConVerse shows that contextually grounded attacks embedded in plausible agent-to-agent discourse are substantially harder than canonical ones; extending that from pairwise conversation to a trust-graded network seems to me the natural next question.

What I bring, and what I do not:

My published work is in a different domain: a super-resolution GAN for photothermal OCT signal enhancement (ICBME 2025, first author), and a dual-wavelength photothermal speckle method for mapping localized absorption in turbid media (under review). What transfers is the experimental half of this problem. My thesis work required building a forward model and an instrument in parallel, establishing ground truth where it is genuinely hard to establish, and characterizing what a method can and cannot resolve before claiming it works, including the unglamorous part, which is finding the conditions under which your own system fails. Benchmark design for contextual integrity has the same shape: the value of the benchmark is determined by how carefully you can control what varies.

I have formal machine learning training, work daily in Python, and have built the deep-learning and simulation pipelines behind my current results independently. The theory of contextual integrity, agent frameworks, and post-training methods are areas I would need to build quickly, and I am prepared for that to be the first six months.

Why this group:

The combination of Dr. Mireshghallah's work grounding LLM privacy in contextual integrity theory and Dr. Abdelnabi's work on indirect prompt injection and agentic security is, as far as I can tell, the closest thing to a complete account of this problem currently assembled in one place. I would like to contribute the empirical and simulation side of it.